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Smoothing spline analysis of variance models: A new tool for the analysis of cyclic biomechanical data

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Abstract

Cyclic biomechanical data are commonplace in orthopedic, rehabilitation, and sports research, where the goal is to understand and compare biomechanical differences between experimental conditions and/or subject populations. A common approach to analyzing cyclic biomechanical data involves averaging the biomechanical signals across cycle replications, and then comparing mean differences at specific points of the cycle. This pointwise analysis approach ignores the functional nature of the data, which can hinder one's ability to find subtle differences between experimental conditions and/or subject populations. To overcome this limitation, we propose using mixed-effects smoothing spline analysis of variance (SSANOVA) to analyze differences in cyclic biomechanical data. The SSANOVA framework makes it possible to decompose the estimated function into the portion that is common across groups (i.e., the average cycle, AC) and the portion that differs across groups (i.e., the contrast cycle, CC). By partitioning the signal in such a manner, we can obtain estimates of the CC differences (CCDs), which are the functions directly describing group differences in the cyclic biomechanical data. Using both simulated and experimental data, we illustrate the benefits of using SSANOVA models to analyze differences in noisy biomechanical (gait) signals collected from multiple locations (joints) of subjects participating in different experimental conditions. Using Bayesian confidence intervals, the SSANOVA results can be used in clinical and research settings to reliably quantify biomechanical differences between experimental conditions and/or subject populations.

Keywords: Cyclic data; Functional data analysis; Gait analysis; Smoothing spline.

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